

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0543

Lirr Island Park 100 kV
February 24, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0543. The Interconnection Customer proposes a 238.4 MW solar photovoltaic facility interconnecting at the Lirr Island Park 100 kV substation in the Illinois area. The request was received on November 28, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 271 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,448,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of BEAR CANYON / CLINTMOORE 100 kV, the NORTH BONNEVILLE SUBSTATION / LAKE CHARLES PLANT 100 kV loads to 282.7% of its 114 MVA rating. Project DFAX: 100.0%.

The N BONNEVILLE / OXBOW 100 kV Line is overloaded to 275.6% of its 116 MVA emergency rating for the loss of the BEAR CANYON / CLINTMOORE SUBSTATION 100 kV. The project contributes 100.0% DFAX (238.4 MW) to this overload.

OXBOW SUBSTATION / BLYTHE 100 kV: 250.2% loading (122 MVA) following the outage of the CLINTMOORE / BEAR CANYON 100 kV Line; DFAX = 100.0%.

SHORELINE / FRONT RANE 100 kV: 241.5% loading (121 MVA) following the outage of the CLINTMOORE / LAKE CHARLES PLANT 100 kV; DFAX = 81.3%.

FRONT RANGE / SOCRATES 19 100 kV: 238.0% loading (123 MVA) following the outage of the CLINTMOORE / LAKE CHARLES PLANT 100 kV; DFAX = 81.3%.

The BEAR CANYON / SOCRATES 19 100 kV is overloaded to 237.9% of its 158 MVA emergency rating for the loss of the LAKE CHARLES PLANT / CLINTMOORE 100 kV. The project contributes 100.0% DFAX (238.4 MW) to this overload.

For the loss of CLINTMOORE / LAKE CHARLES PLANT 100 kV, the BEAR CANYON / CLINTMOORE 100 kV line loads to 220.5% of its 179 MVA rating. Project DFAX: 100.0%.

LAKE CHARLES PLANT / CLINTMOORE 100 kV: 220.5% loading (179 MVA) following the outage of the BEAR CANYON / CLINTMOORE 100 kV; DFAX = 100.0%.

LIRR ISLAND PARK / CASEY W 100 kV: 214.1% loading (131 MVA) following the outage of the BEAVER / EUFAULA SUBSTATION 500 kV; DFAX = 100.0%.

SHORELINE / FLAMBEAU HYDROELECTRIC STA 100 kV: 213.2% loading (129 MVA) following the outage of the CLINTMOORE / LAKE CHARLES PLANT 100 kV Line; DFAX = 81.3%.

CLINTMOORE / CONRAD AUTO 100 kV Line: 213.2% loading (189 MVA) following the outage of the MINNEAPOLIS CITY OF / WALLENPAUPACK SUBSTATION 500 kV; DFAX = 100.0%.

The CASEY WEST / SIERRA BLANCA 100 kV is overloaded to 212.4% of its 130 MVA emergency rating for the loss of the GRAND RIDGE BATTERY PROJECTS SUBSTATION / BREVARD 500 kV. The project contributes 100.0% DFAX (238.4 MW) to this overload.

For the loss of MINNEAPOLIS CITY OF / WALLENPAUPACK 500 kV, the CONRAD AUTO / SIERRA BLANCA 100 kV L loads to 209.5% of its 195 MVA rating. Project DFAX: 100.0%.

For the loss of LAKE CHARLES PLANT / CLINTMORE 100 kV, the CEDAR SUBSTATION / DECATUR ROUTE 51 100 kV loads to 134.6% of its 136 MVA rating. Project DFAX: 19.6%.

DECATUR ROUTE 51 / ASBURY SUBSTATION 100 kV L: 129.5% loading (134 MVA) following the outage of the CLINTMOORE / LAKE CHARLES PLANT 100 kV; DFAX = 19.6%.

The BRANDY BRANCH / ASBURY 100 kV is overloaded to 124.9% of its 131 MVA emergency rating for the loss of the LAKE CHARLES PLANT / CLINTMOORE 100 kV. The project contributes 19.6% DFAX (46.7 MW) to this overload.

BRANDY BRANCH / LEOTA 100 kV line: 119.6% loading (128 MVA) following the outage of the LAKE CHARLES PLANT / CLINTMOORE 100 kV; DFAX = 19.6%.

For the loss of SHOREINE / FRONT RANGE 100 kV, the KENZIE CRK / SOCRATES 19 100 kV loads to 118.3% of its 111 MVA rating. Project DFAX: 41.2%.

For the loss of LAKE CHARLES PLANT / CLINTMOORE 100 kV, the CEDAR / FLAMBEAU HYDROELECTRIC STA 100 kV loads to 118.3% of its 170 MVA rating. Project DFAX: 19.6%.

For the loss of LAWRENCE HILL / SWEATT 100 kV, the SWEATT / TINLEY PARK 100 kV loads to 118.2% of its 132 MVA rating. Project DFAX: 6.3%.

For the loss of CARLYSS / TOP CROP 500 kV, the CARLYSS / BOCA RATON SUBSTATION 230 kV loads to 102.0% of its 560 MVA rating. Project DFAX: 6.9%.

SWEATT / ARCH SMITH 100 kV Circuit 1: 101.3% loading (241 MVA) following the outage of the SWEATT / ARC SMITH SUBSTATION 100 kV L; DFAX = 9.1%.

For the loss of SWEATT / ARCH SMITH 100 kV Line 1, the ACH SMITH / SWEATT SUBSTATION 100 kV loads to 101.3% of its 241 MVA rating. Project DFAX: 9.1%.

The SWEATT / LAWRENCE HILL SUBSTATION 100 kV is overloaded to 101.1% of its 229 MVA emergency rating for the loss of the BLYTHE / S BOSTON 230 kV. The project contributes 8.6% DFAX (20.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N BONNEVILLE / LAKE CHARLES PLANT 100 kV	Rebuild line with higher-capacity conductor	\$3,153,000	62% (\$1,969,000)
N BONNEVILLE / OXBOW 100 kV Line	Rebuild line with higher-capacity conductor	\$8,750,000	62% (\$5,464,000)
OXBOW / BLYTHE 100 kV Line	Rebuild line with higher-capacity conductor	\$1,522,000	62% (\$950,000)
SHORELINE / FRONT RANGE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$800,000	60% (\$479,000)
FRONT RANGE / SOCRATES 19 100 kV	Rebuild line with higher-capacity conductor	\$2,764,000	60% (\$1,655,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEAR CANYON / SOCRATES 19 100 kV	Rebuild line with higher-capacity conductor	\$2,452,000	62% (\$1,531,000)
BEAR CANYON / CLINTMOORE 100 kV line	Rebuild line with higher-capacity conductor	\$4,879,000	62% (\$3,047,000)
CLINTMOORE / LAKE CHARLES PLANT 100 kV Line	Rebuild line with higher-capacity conductor	\$8,925,000	62% (\$5,573,000)
LIRR ISLND PARK / CASEY W 100 kV	Rebuild line with higher-capacity conductor	\$4,127,000	100% (\$4,127,000)
SHORELINE / FLAMBEAU HYDROELECTRIC STA 100 kV	Rebuild line with higher-capacity conductor	\$456,000	60% (\$273,000)
CLINTMOORE / CONRAD AUTO 100 kV	Rebuild line with higher-capacity conductor	\$2,291,000	62% (\$1,431,000)
CASEY WEST / SIERRA BLANCA 100 kV	Rebuild line with higher-capacity conductor	\$10,513,000	100% (\$10,513,000)
SIERRA BLANCA / CONRAD AUTO SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$4,929,000	62% (\$3,078,000)
CEDAR / DECATUR ROUTE 51 100 kV	Rebuild line with higher-capacity conductor	\$1,940,000	42% (\$812,000)
DECATUR ROUTE 51 / ASBURY 100 kV Line	Rebuild line with higher-capacity conductor	\$1,684,000	42% (\$705,000)
BRANDY BRANCH / ASBURY 100 kV	Rebuild line with higher-capacity conductor	\$4,285,000	42% (\$1,794,000)
BRANDY BRANCH / LEOTA 100 kV line	Rebuild line with higher-capacity conductor	\$4,755,000	42% (\$1,991,000)
SOCRATES 19 / KENZIE CREEK 100 kV	Rebuild line with higher-capacity conductor	\$1,959,000	62% (\$1,223,000)
CEDAR / FLAMBEAU HYDROELECTRIC STATION 100 kV Line	Rebuild line with higher-capacity conductor	\$582,000	42% (\$244,000)
TINLEY PARK / SWEATT 100 kV	Rebuild line with higher-capacity conductor	\$9,307,000	24% (\$2,207,000)
CARLYSS / BOCA RATON SUBSTATION 230 kV	Reconductor line	\$22,286,000	22% (\$4,868,000)
SWEATT / ARCH SMITH 100 kV Circuit 1	Reconductor line	\$791,000	22% (\$177,000)
SWEATT SUBSTATION / ARCH SMITH 100 kV #2	Reconductor line	\$507,000	22% (\$114,000)
SWEATT / LAWRENCE HILL SUBSTATION 100 kV	Reconductor line	\$1,228,000	18% (\$223,000)
	Total allocated to project		\$54,448,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.